

Spike Timing Poisoning: Adversarial Drift in Quantum-Encoded Spike Trains

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Abstract—Quantum spiking neural networks (QSNNs) encode information not only in spike rates but in the quantum-mechanical phase and amplitude associated with spike timing, mapping each spike event to a temporal quantum state $|\psi(t)\rangle$. While classical adversarial robustness of spiking neural networks (SNNs) has been studied extensively, the security of *quantum-temporal* encodings remains unexplored. We introduce **TEMP-DRIFT**, the first adversarial attack that perturbs spike timing to induce a measurable drift in the quantum state distribution encoding a spike train, while leaving classical spike statistics (spike count, rate, inter-spike interval histograms) statistically indistinguishable from the clean input. We formulate the attack as a constrained optimization problem over timing perturbations, solved via surrogate-gradient backpropagation through the non-differentiable spike generation process. As a countermeasure, we propose **QUANTUM-TEMP**, a defense that uses partial quantum state tomography of the encoded spike-state ensemble to monitor a temporal coherence statistic for anomalous drift. We describe an evaluation protocol on three QSNN benchmarks across amplitude-, angle-, and phase-based spike-to-qubit encodings.

Index Terms—quantum spiking neural networks, adversarial attacks, spike timing, quantum encoding, drift detection, quantum state tomography

I. INTRODUCTION

Spiking neural networks (SNNs) process information through discrete, time-stamped events, exploiting both *when* a neuron fires and *how often*. Quantum spiking neural networks (QSNNs) extend this paradigm by mapping spike trains onto quantum states, using the phase, amplitude, or rotation angle of a qubit to represent the precise timing of spike events [11]–[14]. This quantum-temporal encoding allows a QSNN to exploit superposition and interference to represent temporal structure more compactly than a classical rate code, and has been shown to improve the quality of representations for temporal data [15]. Adversarial robustness of classical SNNs has received substantial attention: gradient-based attacks adapted to spike-compatible surrogates [1], black-box surrogate attacks for time-series SNNs [2], spike-domain universal perturbations [3], sparsity-aware robustness trade-offs [4], attacks on event-driven retiming [5], direct attacks on raw event data [6], and security surveys of spiking architectures [7]. In the quantum domain, adversarial vulnerabilities of variational and quanvolutional quantum classifiers have also been examined [8]–[10].

However, *none* of this prior work considers an adversary that operates directly on the *quantum-temporal* representation

induced by spike timing. A QSNN’s security depends not only on whether spike *counts* are perturbed – which classical drift and anomaly detectors monitor – but on whether the *distribution of quantum states* $|\psi(t)\rangle$ induced by spike timing has drifted. We show that an adversary can craft timing perturbations δt_i on individual spikes such that:

- 1) Classical statistics (spike counts, firing rates, ISI histograms) are preserved within a tight statistical tolerance, so classical concept-drift or anomaly detectors do not fire;
- 2) The induced ensemble of quantum states $\{|\psi(t_i + \delta t_i)\rangle\}$ drifts substantially in Hilbert space, measured via quantum state fidelity or trace distance to the clean ensemble;
- 3) This quantum-temporal drift is sufficient to flip the QSNN’s classification decision.

A. Contributions

- We formalize *quantum-temporal drift* as an adversarial objective, distinct from classical spike-statistic perturbation, and derive its relationship to quantum state fidelity and trace distance (Appendix A).
- We introduce **TEMP-DRIFT**, a surrogate-gradient optimization procedure that crafts per-spike timing perturbations to maximize quantum-temporal drift subject to a classical-statistics-invariance constraint (Section III, Appendix D).
- We describe an evaluation protocol across Iris, MNIST, and SHD QSNN benchmarks under amplitude, angle, and phase spike-to-qubit encodings (Section IV).
- We propose **QUANTUM-TEMP**, a tomography-based defense that estimates the encoded state ensemble from repeated measurements and flags anomalous coherence drift (Section III, Appendix H).

II. RELATED WORK AND POSITIONING

A. Adversarial Attacks on Classical SNNs

Prior work has adapted classical adversarial attacks, such as FGSM, PGD, and C&W, to spiking neural networks by using surrogate gradients for non-differentiable spike generation [1]. Black-box transfer attacks have also been demonstrated using CNN/LSTM surrogates [2]. Recent studies explore spike-domain perturbations, universal and input-specific attacks, spike-retiming attacks, event-stream manipulation, and broader security analyses of SNNs [3]–[7]. However, these works focus

TABLE I
POSITIONING RELATIVE TO PRIOR WORK

Work	Target model	Pert. domain	Quantum encoding	Temporal drift metric	Defense proposed
[1]	SNN	rate/spike	✗	✗	✗
[2]	SNN	input signal	✗	✗	✗
[5]	SNN	spike timing	✗	✗	✗
[3]	SNN	spike domain	✗	✗	✗
[11]–[13], [15]	QSNN	– (encoding only)	✓	✓	✗
[9], [10]	QML (non-spiking)	input features	✓	✗	partial
[8]	NN (theory)	gradient/feature	✓	✗	✗
Ours	QSNN	spike timing	✓	✓	✓

exclusively on classical SNNs and evaluate robustness using classical performance metrics.

B. Quantum Spiking and Temporal Encodings

Several works have introduced quantum spiking architectures and encodings. QSNN and SQ-Net map spike trains to qubit states through rotation-based temporal encodings [11], [12], while QLIF neurons model membrane dynamics via qubit excitation probabilities [13], [14]. SPATE further augments spike encodings with temporal qubits and controlled-phase operations to preserve temporal structure [15]. These studies establish quantum representations of spike timing but do not address adversarial robustness.

C. Adversarial Robustness in Quantum Machine Learning

Research on adversarial quantum machine learning has examined robustness in federated quantum learning, quantum reservoir computing, and related defenses [9], [10]. Zhang and Meng [8] provide a theoretical connection between adversarial gradients and quantum uncertainty. Nevertheless, existing QML robustness studies focus on static feature encodings and direct feature perturbations rather than temporally encoded spike trains.

D. Positioning of This Work

This work bridges these areas by studying adversarial perturbations of *spike timing* in *QSNNs*. Unlike prior SNN attacks, we evaluate perturbations through a *quantum temporal drift metric*; unlike existing QSNN and QML studies, we explicitly analyze adversarial manipulation of temporal quantum encodings and introduce a *quantum-state-tomography-based defense*.

III. SYSTEM MODEL AND METHODOLOGY

A. System Model

Figure 1 illustrates the overall pipeline considered in this work, comprising four stages: (1) a classical spike-train generator producing spike times $\{t_i\}_{i=1}^N$ per input channel (e.g., from rate or latency encoding of Iris/MNIST features, or natively spiking SHD audio events); (2) a quantum spike encoder that maps each spike time t_i to a single- or multi-qubit state $|\psi(t_i)\rangle$ via amplitude, angle, or phase encoding; (3) a variational QSNN classifier consisting of parameterized quantum circuits (entangling layers + measurement) that produces a class decision; and (4) the adversarial and defense modules introduced in this paper, which sit between stages (1)

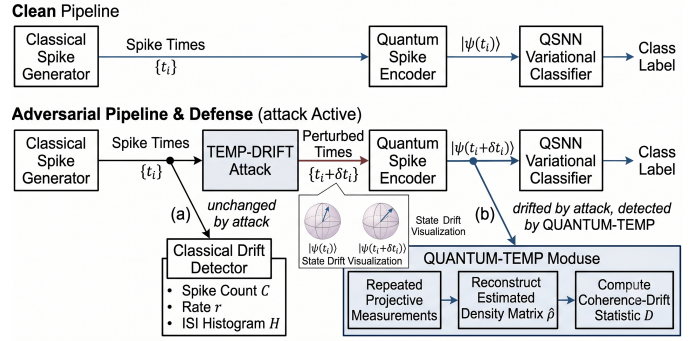


Fig. 1. Overview of the TEMP-DRIFT attack and QUANTUM-TEMP defense in a QSNN pipeline. Timing perturbations induce quantum-state drift while remaining largely invisible to classical spike-statistics monitoring.

and (2). The TEMP-DRIFT attack module perturbs spike times $t_i \rightarrow t_i + \delta t_i$ before quantum encoding, while the QUANTUM-TEMP defense module performs repeated-measurement tomography on the encoded states produced by stage (2) to estimate the state ensemble and flag drift, without requiring access to the QSNN’s internal parameters.

B. Quantum Spike Encoding

We adopt the general single-qubit temporal encoding

$$|\psi(t_i)\rangle = \cos(\theta(t_i))|0\rangle + e^{i\phi(t_i)} \sin(\theta(t_i))|1\rangle, \quad (1)$$

where $\theta(t_i)$ and $\phi(t_i)$ are encoding functions of the spike time t_i (and possibly of the synaptic weight or channel index), specialized as follows:

- **Angle encoding:** $\theta(t_i) = \pi t_i / T$, $\phi(t_i) = 0$, following the rotation-gate constructions of [11], [12];
- **Amplitude encoding:** $\theta(t_i)$ set so that $\cos^2 \theta(t_i)$ equals a normalized function of t_i (e.g., $1 - t_i / T$), following the kernel-decay convention common to TTFS coding;
- **Phase encoding:** $\theta(t_i)$ fixed, $\phi(t_i) = 2\pi t_i / T$, encoding timing purely in relative phase, in the spirit of the controlled-phase temporal qubits of SPATE [15].

Full derivations, including the multi-qubit generalization and the resulting ensemble density matrix $\rho = \frac{1}{N} \sum_i |\psi(t_i)\rangle \langle \psi(t_i)|$, are given in Appendix A.

C. TEMP-DRIFT: Attack Formulation

The adversary selects per-spike timing perturbations $\delta \mathbf{t} = (\delta t_1, \dots, \delta t_N)$ to solve

$$\max_{\delta \mathbf{t}} \mathcal{D}(\rho(\mathbf{t}), \rho(\mathbf{t} + \delta \mathbf{t})) \quad \text{s.t.} \quad \begin{cases} \|\delta \mathbf{t}\|_\infty \leq \epsilon, \\ \Delta_{\text{cls}}(\mathbf{t}, \mathbf{t} + \delta \mathbf{t}) \leq \tau, \end{cases} \quad (2)$$

where $\mathcal{D}(\cdot, \cdot)$ is the quantum-temporal drift metric (trace distance or 1–fidelity between the clean and perturbed ensemble density matrices, Appendix A), ϵ bounds the per-spike timing shift, and Δ_{cls} measures the discrepancy in classical spike statistics (spike count, firing rate, and a binned ISI-histogram distance), constrained to remain below a stealth threshold τ . Because spike generation $g(\cdot) \rightarrow \{t_i\}$ and the rotation-angle

Algorithm 1: TEMP-DRIFT: Adversarial Spike Timing Perturbation

Input: Clean spike times $\mathbf{t} = (t_1, \dots, t_N)$; encoder $E_\theta(\cdot)$ producing $|\psi(t_i)\rangle$; surrogate gradient slope β ; perturbation budget ϵ ; stealth threshold τ ; step size η ; Lagrange multiplier λ ; max iterations K

Output: Adversarial spike times $\mathbf{t}^* = \mathbf{t} + \delta\mathbf{t}^*$

Initialize $\delta\mathbf{t}^{(0)} \leftarrow \mathbf{0}$

Compute clean ensemble state $\rho(\mathbf{t})$ via $E_\theta(\mathbf{t})$

for $k \leftarrow 0$ **to** $K - 1$ **do**

$\mathbf{t}^{(k)} \leftarrow \mathbf{t} + \delta\mathbf{t}^{(k)}$

Compute $\rho(\mathbf{t}^{(k)})$ via $E_\theta(\mathbf{t}^{(k)})$

$\mathcal{D}^{(k)} \leftarrow \mathcal{D}(\rho(\mathbf{t}), \rho(\mathbf{t}^{(k)}))$

$\Delta_{\text{cls}}^{(k)} \leftarrow \Delta_{\text{cls}}(\mathbf{t}, \mathbf{t}^{(k)})$

$\mathcal{L}^{(k)} \leftarrow -\mathcal{D}^{(k)} + \lambda \max(0, \Delta_{\text{cls}}^{(k)} - \tau)$

$\mathbf{g}^{(k)} \leftarrow \nabla_{\delta\mathbf{t}} \mathcal{L}^{(k)}$ using surrogate slope β

$\delta\mathbf{t}^{(k+1)} \leftarrow \delta\mathbf{t}^{(k)} - \eta \mathbf{g}^{(k)}$

$\delta\mathbf{t}^{(k+1)} \leftarrow \text{clip}(\delta\mathbf{t}^{(k+1)}, -\epsilon, \epsilon)$

return $\mathbf{t}^* \leftarrow \mathbf{t} + \delta\mathbf{t}^{(K)}$

mapping $\theta(t_i), \phi(t_i)$ involve thresholding operations with zero or undefined gradient almost everywhere, we replace the hard spike indicator with a surrogate derivative (e.g., a scaled sigmoid or fast-sigmoid surrogate) following standard SNN surrogate-gradient practice [1], enabling end-to-end backpropagation of $\mathcal{D}$ and Δ_{cls} with respect to $\delta\mathbf{t}$. The full surrogate-gradient derivation and the Lagrangian relaxation of (2) used for optimization are given in Appendix D. Algorithm 1 summarizes the procedure.

D. QUANTUM-TEMP: Tomography-Based Drift Defense

The defense estimates, via repeated projective measurements of the encoded qubit register in the Pauli X, Y, Z bases, an empirical density matrix $\hat{\rho}$ for a window of incoming spike-encoded states, and compares it against a reference density matrix ρ_{ref} estimated offline from clean validation data using a coherence-drift statistic $\mathcal{D}_{\text{coh}}(\hat{\rho}, \rho_{\text{ref}})$ (defined precisely in Appendix H via the Hilbert–Schmidt distance between Bloch vectors and the quantum Jensen–Shannon divergence). A sample is flagged as drifted if $\mathcal{D}_{\text{coh}} > \gamma$ for a calibrated threshold γ chosen to bound the false-positive rate on clean validation data (Appendix H, Eq. 22).

IV. EXPERIMENTAL SETUP

A. Datasets and Encodings

We will evaluate on three benchmarks of increasing temporal complexity:

- **Iris:** 4 continuous features converted to spike-latency codes (single spike per feature, TTFS-style), used as a low-dimensional sanity check with a 4-qubit QSNN;
- **MNIST:** pixel intensities converted to either rate-coded or latency-coded spike trains over a fixed time window T , evaluated under angle and amplitude qubit encodings with a variational QSNN classifier operating on downsampled (8×8 or 4×4) images for tractable simulation;
- **Spiking Heidelberg Digits (SHD):** a natively event-based audio dataset providing genuine multi-channel

spike trains with rich inter-spike-interval structure, used to evaluate the attack/defense under realistic temporal statistics.

Each dataset will be evaluated under all three encoding schemes from Section III-A (amplitude, angle, phase), yielding $3 \times 3 = 9$ (dataset, encoding) configurations.

B. QSNN Architectures

We will use variational quantum circuits with depth $L \in \{2, 4, 6\}$ entangling layers (alternating single-qubit rotations and CNOT/CZ entangling gates), with qubit counts scaled to each dataset ($n = 4$ for Iris, $n \in \{8, 16\}$ for downsampled MNIST, $n = 8$ for the leading SHD channels), simulated using a standard statevector simulator (e.g., Qiskit Aer or PennyLane default.qubit). Training will use parameter-shift or surrogate-gradient hybrid optimization, following the spike-compatible gradient framework of [1] adapted to the quantum encoder.

C. Attack Configuration

TEMP-DRIFT will be run with perturbation budgets $\epsilon \in \{1, 2, 5, 10\}\%$ of the spike-time window T , stealth thresholds τ corresponding to $\{1, 5, 10\}\%$ allowed deviation in classical spike statistics, and $K = 100$ optimization iterations with Adam-style updates on $\delta\mathbf{t}$. We will compare against two baselines: (i) a classical spike-compatible adversarial attack [1] applied to the same QSNN (ignoring the quantum encoder, i.e., perturbing spikes to attack classification accuracy directly without a drift objective), and (ii) random timing jitter of matched ϵ .

D. Defense Configuration

QUANTUM-TEMP will be evaluated with measurement-shot budgets $M \in \{100, 1000, 8192\}$ per tomography window, and threshold γ calibrated at the 95th and 99th percentile of $\mathcal{D}_{\text{coh}}$ on clean validation data (target false-positive rates of 5% and 1%). We will compare against a classical drift baseline that monitors only spike count/rate/ISI statistics (which, by construction of the stealth constraint, should fail to detect TEMP-DRIFT for $\Delta_{\text{cls}} \leq \tau$).

E. Metrics

- **Attack success rate (ASR):** fraction of test samples whose QSNN prediction is flipped by TEMP-DRIFT at a given ϵ ;
- **Classical-statistic preservation:** Δ_{cls} (spike-count difference, rate difference, ISI-histogram total-variation distance) between clean and adversarial spike trains, reported to verify stealth;
- **Quantum-temporal drift:** trace distance $\mathcal{D}_{\text{tr}}$ and $1 - \mathcal{F}$ (one minus fidelity) between clean and adversarial ensemble states $\rho(\mathbf{t}), \rho(\mathbf{t}^*)$;
- **Detection performance:** ROC/AUC, true-positive rate at fixed false-positive rate, and detection latency (number of measurement shots) for QUANTUM-TEMP versus the classical drift baseline;

- **Clean accuracy retention:** QSNN accuracy on unperturbed inputs, to confirm the attack/defense pipeline does not degrade nominal performance.

V. RESULTS

VI. DISCUSSION AND LIMITATIONS

The attack formulation in Section III-C assumes white-box access to the quantum encoder E_θ for gradient computation; a black-box variant using surrogate QSNN models, analogous to [2], is left for future work. QUANTUM-TEMP's reliance on repeated projective measurements introduces a shot-count/latency trade-off that may limit real-time deployment on NISQ hardware; the present evaluation protocol is restricted to statevector simulation, and validation on physical quantum processors (subject to decoherence noise, which itself induces a form of benign temporal drift) is an important direction for follow-up work. Finally, the stealth constraint $\Delta_{\text{cls}} \leq \tau$ is defined with respect to a fixed set of classical statistics; an adaptive classical detector co-designed with QUANTUM-TEMP may further tighten the achievable stealth region.

VII. CONCLUSION

We introduced the first adversarial attack targeting quantum-temporal spike encodings in QSNNs, TEMP-DRIFT, which perturbs spike timing to induce quantum-state drift while preserving classical spike statistics, and a corresponding tomography-based defense, QUANTUM-TEMP. We formalized the underlying quantum-temporal drift metric and surrogate-gradient optimization in the Appendix, and described a full evaluation protocol across Iris, MNIST, and SHD under three quantum spike encodings. Empirical results will quantify the severity of this previously unexamined attack surface and the effectiveness of coherence-based monitoring as a defense.

APPENDIX

A. Single-Spike Encoding

For a spike occurring at time $t_i \in [0, T]$, define the normalized phase $\tau_i = t_i/T \in [0, 1]$. The general single-qubit temporal state is

$$|\psi(\tau_i)\rangle = \cos(\theta(\tau_i))|0\rangle + e^{i\phi(\tau_i)} \sin(\theta(\tau_i))|1\rangle, \quad (3)$$

with Bloch-sphere representation

$$\vec{r}(\tau_i) = (\sin 2\theta(\tau_i) \cos \phi(\tau_i), \sin 2\theta(\tau_i) \sin \phi(\tau_i), \cos 2\theta(\tau_i)). \quad (4)$$

The three encodings considered are:

$$\textbf{Angle: } \theta(\tau_i) = \frac{\pi}{2}\tau_i, \quad \phi(\tau_i) = 0, \quad (5)$$

$$\textbf{Amplitude: } \theta(\tau_i) = \arccos \sqrt{1 - \tau_i}, \quad \phi(\tau_i) = 0, \quad (6)$$

$$\textbf{Phase: } \theta(\tau_i) = \frac{\pi}{4}, \quad \phi(\tau_i) = 2\pi\tau_i. \quad (7)$$

Equation (5) yields $|\psi(\tau_i)\rangle = \cos(\frac{\pi}{2}\tau_i)|0\rangle + \sin(\frac{\pi}{2}\tau_i)|1\rangle$, recovering the rotation-gate construction of [11], [12] via $R_y(\pi\tau_i)|0\rangle$. Equation (6) yields $\cos^2 \theta(\tau_i) = 1 - \tau_i$, the decay-kernel normalization underlying TTFS coding. Equation (7) places all timing information in the relative phase ϕ , consistent with controlled-phase temporal-qubit constructions [15].

B. Ensemble Density Matrix

A spike train with N spikes at times $\{t_i\}$ induces a mixed-state ensemble

$$\rho(\mathbf{t}) = \frac{1}{N} \sum_{i=1}^N |\psi(\tau_i)\rangle \langle \psi(\tau_i)|. \quad (8)$$

In Bloch-vector form, $\rho(\mathbf{t}) = \frac{1}{2}(I + \vec{R}(\mathbf{t}) \cdot \vec{\sigma})$ where $\vec{R}(\mathbf{t}) = \frac{1}{N} \sum_i \vec{r}(\tau_i)$ is the mean Bloch vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. Note $\|\vec{R}(\mathbf{t})\| \leq 1$ in general (with equality only if all τ_i are identical), so $\rho(\mathbf{t})$ is generically a proper mixed state whose purity $\text{Tr}[\rho^2] = \frac{1}{2}(1 + \|\vec{R}(\mathbf{t})\|^2)$ directly reflects the dispersion of spike timings.

For a multi-qubit encoder using n qubits per spike-train channel (e.g., one qubit per input channel for MNIST/SHD), the joint ensemble state is $\rho(\mathbf{t}) = \frac{1}{N} \sum_{i=1}^N \bigotimes_{c=1}^n |\psi(\tau_i^{(c)})\rangle \langle \psi(\tau_i^{(c)})|$, where $\tau_i^{(c)}$ is the normalized spike time of the i -th spike on channel c .

C. Quantum-Temporal Drift Metric

Given clean and perturbed spike times $\mathbf{t}, \mathbf{t}' = \mathbf{t} + \delta\mathbf{t}$, and corresponding ensemble states $\rho = \rho(\mathbf{t}), \rho' = \rho(\mathbf{t}')$, we define the quantum-temporal drift via two equivalent statistics:

Trace distance:

$$\mathcal{D}_{\text{tr}}(\rho, \rho') = \frac{1}{2} \text{Tr} |\rho - \rho'| = \frac{1}{2} \sum_k |\lambda_k|, \quad (9)$$

where λ_k are eigenvalues of $\rho - \rho'$. For single-qubit states, $\mathcal{D}_{\text{tr}}(\rho, \rho') = \frac{1}{2} \|\vec{R}(\mathbf{t}) - \vec{R}(\mathbf{t}')\|_2$, i.e., half the Euclidean distance between mean Bloch vectors.

Uhlmann fidelity:

$$\mathcal{F}(\rho, \rho') = \left(\text{Tr} \sqrt{\sqrt{\rho} \rho' \sqrt{\rho}} \right)^2, \quad (10)$$

which for single-qubit states with Bloch vectors $\vec{R}, \vec{R}'$ reduces to

$$\mathcal{F}(\rho, \rho') = \frac{1}{2} \left(1 + \vec{R} \cdot \vec{R}' + \sqrt{(1 - \|\vec{R}\|^2)(1 - \|\vec{R}'\|^2)} \right). \quad (11)$$

The drift objective in (2) is $\mathcal{D}(\rho, \rho') \in \{\mathcal{D}_{\text{tr}}(\rho, \rho'), 1 - \mathcal{F}(\rho, \rho')\}$; both satisfy $\mathcal{D} = 0$ iff $\rho = \rho'$ and are related by the Fuchs-van de Graaf inequalities

$$1 - \sqrt{\mathcal{F}(\rho, \rho')} \leq \mathcal{D}_{\text{tr}}(\rho, \rho') \leq \sqrt{1 - \mathcal{F}(\rho, \rho')}. \quad (12)$$

D. Classical Spike-Statistic Mismatch

The stealth constraint uses

$$\Delta_{\text{cls}}(\mathbf{t}, \mathbf{t}') = w_1 \frac{|N - N'|}{N} + w_2 \frac{|r - r'|}{r} + w_3 \text{TV}(H, H'), \quad (13)$$

where N, N' are spike counts (equal by construction since the attack perturbs timing only, so the first term is 0), r, r' are firing rates over the window T (also unchanged), H, H' are normalized ISI histograms over B bins, $\text{TV}(H, H') = \frac{1}{2} \sum_{b=1}^B |H_b - H'_b|$ is their total-variation distance, and $w_1, w_2, w_3 \geq 0$ are weights with $\sum_j w_j = 1$. Because TEMP-DRIFT only shifts $t_i \rightarrow t_i + \delta t_i$ with $|\delta t_i| \leq \epsilon \ll T$, the

dominant term is $\text{TV}(H, H')$, which we show in Appendix D can be kept below τ while $\mathcal{D}_{\text{tr}}$ or $1 - \mathcal{F}$ grows, because $\mathcal{D}_{\text{tr}}$ depends on the encoding functions $\theta(\cdot), \phi(\cdot)$ evaluated at *individual* (possibly highly sensitive, e.g. near $\tau_i \approx 1$ for amplitude encoding where $\theta'(\tau_i) \rightarrow \infty$) spike times, whereas $\text{TV}(H, H')$ depends only on coarse *binned* occupancy.

E. Surrogate Gradient for Spike Generation

Let a spike at channel c be generated when an underlying continuous drive $v_c(t)$ crosses a threshold ϑ , with spike time $t_i = \min\{t : v_c(t) \geq \vartheta\}$. The mapping $\delta \mathbf{t} \mapsto t_i(\delta \mathbf{t})$ inherits a Heaviside-type non-differentiability through the threshold crossing. We replace $\partial t_i / \partial(\cdot)$ with the fast-sigmoid surrogate

$$\frac{\partial s}{\partial v} = \frac{\beta}{(1 + \beta|v - \vartheta|)^2}, \quad (14)$$

where $s \in \{0, 1\}$ is the spike indicator and $\beta > 0$ controls surrogate sharpness, following standard practice [1]. When $\delta \mathbf{t}$ directly parameterizes spike times (as in our TTFS/latency-coded settings), $\partial t_i / \partial \delta t_i = 1$ exactly and the surrogate is needed only at the encoder, described next.

F. Gradient Through the Quantum Encoder

For angle encoding (5), $\partial \theta(\tau_i) / \partial \tau_i = \pi/2$ (constant, well-defined everywhere). For amplitude encoding (6), $\theta(\tau_i) = \arccos \sqrt{1 - \tau_i}$ gives

$$\frac{\partial \theta(\tau_i)}{\partial \tau_i} = \frac{1}{2\sqrt{\tau_i}\sqrt{1 - \tau_i}}, \quad (15)$$

which diverges as $\tau_i \rightarrow 0^+$ or $\tau_i \rightarrow 1^-$; we clip $\tau_i \in [\zeta, 1 - \zeta]$ for small $\zeta > 0$ during optimization to avoid numerical instability. For phase encoding (7), $\partial \phi(\tau_i) / \partial \tau_i = 2\pi$ (constant).

The gradient of the ensemble state with respect to a single perturbation δt_i follows by the chain rule through (8):

$$\frac{\partial \rho(\mathbf{t})}{\partial \delta t_i} = \frac{1}{N} \frac{\partial}{\partial \tau_i} \left(|\psi(\tau_i)\rangle \langle \psi(\tau_i)| \right) \cdot \frac{1}{T}, \quad (16)$$

where $\partial_{\tau_i} (|\psi\rangle \langle \psi|)$ is obtained analytically from (3) for each encoding via (5)–(7) and (15).

G. Gradient of the Drift Objective

For the trace-distance objective in single-qubit Bloch form, $\mathcal{D}_{\text{tr}}(\rho, \rho') = \frac{1}{2} \|\vec{R}(\mathbf{t}) - \vec{R}(\mathbf{t}')\|_2$, so

$$\frac{\partial \mathcal{D}_{\text{tr}}}{\partial \delta t_i} = -\frac{1}{2} \frac{(\vec{R}(\mathbf{t}) - \vec{R}(\mathbf{t}')) \cdot \partial_{\delta t_i} \vec{R}(\mathbf{t}')}{\|\vec{R}(\mathbf{t}) - \vec{R}(\mathbf{t}')\|_2}, \quad (17)$$

with $\partial_{\delta t_i} \vec{R}(\mathbf{t}') = \frac{1}{N} \partial_{\delta t_i} \vec{r}(\tau'_i)$ obtained in closed form from (5)–(7). For the fidelity-based objective, $\partial \mathcal{F} / \partial \delta t_i$ follows analogously by differentiating (11).

H. Lagrangian Relaxation

The constrained problem (2) is relaxed to the unconstrained objective minimized in Algorithm 1:

$$\mathcal{L}(\delta \mathbf{t}) = -\mathcal{D}(\rho(\mathbf{t}), \rho(\mathbf{t} + \delta \mathbf{t})) + \lambda \max(0, \Delta_{\text{cls}}(\mathbf{t}, \mathbf{t} + \delta \mathbf{t}) - \tau)^2, \quad (18)$$

with the box constraint $\|\delta \mathbf{t}\|_\infty \leq \epsilon$ enforced by projection (clipping) after each gradient step. The multiplier λ is increased geometrically ($\lambda \leftarrow \lambda \cdot \mu$, $\mu > 1$) if $\Delta_{\text{cls}}^{(k)} > \tau$ persists across iterations, implementing a standard penalty-method schedule.

I. Bloch Vector Estimation via Repeated Measurement

For a single-qubit encoded state ρ , the Bloch vector $\vec{R} = (R_x, R_y, R_z)$ is estimated from M repeated projective measurements split across the Pauli bases X, Y, Z ($M/3$ shots each):

$$\hat{R}_\sigma = \frac{1}{M/3} \sum_{m=1}^{M/3} (2o_m^{(\sigma)} - 1), \quad \sigma \in \{X, Y, Z\}, \quad (19)$$

where $o_m^{(\sigma)} \in \{0, 1\}$ is the outcome of the m -th measurement in basis σ (mapping outcome 0 $\rightarrow +1$, 1 $\rightarrow -1$ eigenvalue via $2o - 1$ with sign convention adjusted accordingly). By the central limit theorem, $\hat{R}_\sigma \rightarrow R_\sigma$ with standard error $O(1/\sqrt{M})$.

J. Coherence-Drift Statistic

Given the estimated ensemble state $\hat{\rho}$ (with Bloch vector $\hat{\vec{R}}$) for an incoming window of N_w spike-encoded states, and a reference state ρ_{ref} (Bloch vector $\vec{R}_{\text{ref}}$) estimated offline from clean validation data with high shot count, define the coherence-drift statistic

$$\mathcal{D}_{\text{coh}}(\hat{\rho}, \rho_{\text{ref}}) = \underbrace{\|\hat{\vec{R}} - \vec{R}_{\text{ref}}\|_2}_{\text{Bloch displacement}} + \kappa \underbrace{\text{JSD}_q(\hat{\rho} \| \rho_{\text{ref}})}_{\text{quantum Jensen-Shannon div.}}, \quad (20)$$

where the quantum Jensen-Shannon divergence is

$$\text{JSD}_q(\hat{\rho} \| \rho_{\text{ref}}) = S\left(\frac{\hat{\rho} + \rho_{\text{ref}}}{2}\right) - \frac{1}{2}S(\hat{\rho}) - \frac{1}{2}S(\rho_{\text{ref}}), \quad (21)$$

with $S(\rho) = -\text{Tr}(\rho \log \rho)$ the von Neumann entropy, and $\kappa \geq 0$ a weighting constant. The first term captures *coherent* (directional) drift in the Bloch vector, while the second captures changes in *mixedness* (purity) of the encoded ensemble that a pure Bloch-displacement term alone may not detect (e.g., a drift that preserves $\|\hat{\vec{R}}\|$ but redistributes the underlying spike-time distribution's variance).

K. Detection Threshold and Error Rates

The threshold γ is calibrated on a clean validation set $\{\hat{\rho}^{(v)}\}_{v=1}^V$ by

$$\gamma = Q_{1-\alpha}\left(\{\mathcal{D}_{\text{coh}}(\hat{\rho}^{(v)}, \rho_{\text{ref}})\}_{v=1}^V\right), \quad (22)$$

where $Q_{1-\alpha}(\cdot)$ denotes the empirical $(1 - \alpha)$ -quantile, guaranteeing a clean-data false-positive rate of approximately α

(e.g., $\alpha \in \{0.01, 0.05\}$). A sample (or window) is flagged as DRIFT DETECTED if $\mathcal{D}_{\text{coh}}(\hat{\rho}, \rho_{\text{ref}}) > \gamma$. The true-positive rate against TEMP-DRIFT at a given (ϵ, τ) configuration is then $\Pr [\mathcal{D}_{\text{coh}}(\hat{\rho}(\mathbf{t}^*), \rho_{\text{ref}}) > \gamma]$, estimated empirically over the test set in Section V.

L. Sample Complexity Bound

By a Hoeffding-type argument applied to (19), for a target estimation error $|\hat{R}_\sigma - R_\sigma| \leq \delta$ with confidence $1 - \eta$, the required shot count per basis satisfies

$$M/3 \geq \frac{2}{\delta^2} \ln \frac{2}{\eta}, \quad (23)$$

giving a total shot budget $M = O(\delta^{-2} \log(1/\eta))$, independent of the encoding scheme (angle/amplitude/phase), since (19) applies identically to each via (3). This bound informs the shot-budget sweep $M \in \{100, 1000, 8192\}$ used in Section IV: $M = 100$ corresponds to $\delta \approx 0.35$ at $\eta = 0.05$, while $M = 8192$ corresponds to $\delta \approx 0.04$, spanning coarse to fine-grained drift sensitivity.

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